Teacher Wu Kuan | Exam Trap — Counting Terms in an Arithmetic Sequence

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1. The trap question

3 + 7 + 11 + … + 99: how many terms? Many students answer 24.

2. The method

• (last − first) ÷ common difference counts the GAPS between terms.
• Number of terms = gaps + 1 (the “fence-post” / tree-planting rule).
• Here: (99 − 3) ÷ 4 = 24 gaps → 25 terms.
• Sum = number of terms × (first + last) ÷ 2 = 25 × 102 ÷ 2 = 1275.

3. Parent check method

Ask: “If you put 3 fence posts, how many gaps are there?” If your child says 2, they understand why we add 1.

Practice

1. How many terms are in 5, 8, 11, …, 98?
2. Find 2 + 6 + 10 + … + 98.
3. How many terms are in 10, 15, 20, …, 200?
4. A 60 m fence has a post every 5 m, including both ends. How many posts?
5. Find 1 + 4 + 7 + … + 100.

Answers

1. 32 ((98 − 5) ÷ 3 = 31 gaps, +1)
2. 1250 (25 terms; 25 × 100 ÷ 2)
3. 39 (190 ÷ 5 = 38 gaps, +1)
4. 13
5. 1717 (34 terms; 34 × 101 ÷ 2)

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